

APPENDICES OF CLASSIFYING INTEGRAL GROTHENDIECK RINGS UP TO RANK 5 AND BEYOND

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Appendix A. INTEGRAL DRINFELD RINGS

A.1. Up to rank 5. This section presents the comprehensive list of integral Drinfeld rings up to rank 5—including their global FPdim, type, duality, formal codegrees, and fusion data. Copy-pastable data can be found in the file `GeneralUpToRank5DataOnly.txt`, located in the `Data/General` directory of [7]. For each case, either an explicit categorification is provided, or a reference is given to a theoretical result ruling out its existence.

A.1.1. *Rank 1.* Trivial case

A.1.2. *Rank 2.*

(1) FPdim 2, type $[1, 1]$, duality $[0, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[2, 2]$,
- Property: simple,
- Categorification: $\text{Rep}(C_2)$.

A.1.3. *Rank 3.*

(1) FPdim 3, type $[1, 1, 1]$, duality $[0, 2, 1]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 3]$,
- Property: simple,
- Categorification: $\text{Rep}(C_3)$.

(2) FPdim 6, type $[1, 1, 2]$, duality $[0, 1, 2]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

- Formal codegrees: $[2, 3, 7, 43, 1806]$,
- Categorification: excluded by the fusion subring of type $[1, 1, 2, 6]$, see §A.1.4 (5).

- Formal codegrees: $[4, 5, 5, 7, 7, 16, 560]$,
- Property: simple, non-1-Frobenius,
- Categorification: open, non-braided.

(3) FPdim 2860, type $[1, 11, 12, 12, 15, 25, 40]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 2 \\ 0 & 0 & 1 & 0 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 2 \\ 1 & 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 0 & 1 & 1 & 1 & 2 \\ 0 & 0 & 1 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 1 & 1 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 0 & 1 & 1 & 1 & 1 & 2 & 2 \\ 1 & 0 & 1 & 1 & 3 & 3 & 2 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 2 & 4 & 10 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 1 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 1 & 2 & 3 & 3 & 4 & 6 & 8 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 2 & 2 & 2 & 2 & 4 & 6 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 7 \\ 0 & 2 & 2 & 2 & 2 & 4 & 10 \\ 0 & 4 & 4 & 4 & 4 & 8 & 15 \\ 1 & 6 & 7 & 7 & 10 & 15 & 21 \end{bmatrix}$$

- Formal codegrees: $[4, 4, 5, 5, 13, 44, 2860]$,
- Property: simple, non-1-Frobenius,
- Categorification: open, non-braided.

(4) FPdim 3192, type $[1, 15, 16, 17, 18, 24, 39]$, duality $[0, 1, 2, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 3 & 3 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 2 & 4 & 4 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 1 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 1 & 1 & 2 & 1 & 3 & 3 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 1 & 2 & 4 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 & 2 & 2 & 4 \\ 0 & 1 & 1 & 1 & 3 & 2 & 3 \\ 0 & 2 & 2 & 2 & 3 & 2 & 6 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 3 & 3 & 3 & 2 & 1 & 4 \\ 0 & 3 & 3 & 3 & 2 & 2 & 4 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 3 & 3 & 3 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 3 & 2 & 6 \\ 0 & 4 & 4 & 4 & 6 & 7 & 12 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 3 & 3 & 4 & 4 & 7 \\ 0 & 3 & 3 & 3 & 4 & 4 & 8 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 3 & 3 & 3 & 4 & 4 & 9 \\ 0 & 4 & 4 & 4 & 3 & 6 & 8 \\ 0 & 4 & 4 & 4 & 6 & 7 & 12 \\ 1 & 7 & 8 & 9 & 8 & 12 & 18 \end{bmatrix}$$

- Formal codegrees: $[3, 3, 6, 8, 42, 57, 3192]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification: open, non-braided, non-unitary

(5) FPdim 4284, type $[1, 17, 17, 20, 28, 35, 36]$, duality $[0, 2, 1, 3, 4, 5, 6]$, fusion data:

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 2 & 2 & 2 \\ 0 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 2 & 2 & 3 \\ 0 & 2 & 1 & 2 & 2 & 2 & 2 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 1 & 2 & 3 & 3 \\ 0 & 2 & 1 & 1 & 2 & 3 & 3 \\ 0 & 1 & 1 & 1 & 3 & 2 & 3 & 4 \\ 0 & 2 & 2 & 2 & 6 & 4 & 4 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 3 & 4 & 4 \\ 0 & 2 & 2 & 2 & 6 & 4 & 4 \\ 1 & 3 & 3 & 6 & 1 & 7 & 8 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 2 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 3 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 7 & 8 & 8 \\ 1 & 5 & 5 & 6 & 8 & 10 & 10 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 3 & 2 & 3 & 4 & 5 & 5 \\ 0 & 2 & 3 & 3 & 4 & 5 & 5 \\ 0 & 3 & 3 & 4 & 4 & 6 & 6 \\ 0 & 4 & 4 & 4 & 8 & 8 & 8 \\ 0 & 5 & 5 & 6 & 8 & 10 & 11 \\ 1 & 5 & 5 & 6 & 8 & 10 & 11 \end{bmatrix}$$

- Formal codegrees: $[3, 4, 7, 7, 9, 51, 4284]$,
- Property: simple, non-1-Frobenius, non-3-positive
- Categorification: open, non-braided, non-unitary

These fusion rings also invite questions in the spirit of Question 1.13.

A.4. Ranks 8 and 9. There are 792 integral 1-Frobenius Drinfeld rings of rank 8 with $\text{FPdim} \leq 25000$, and 1292 such rings of rank 9 with $\text{FPdim} \leq 2000$. Copy-pastable data can be found in the files `1FrobR8d25000.txt` and `1FrobR9d2000.txt`, located in the `Data/General` directory of [7].

A.5. Addressing divisibility. The classification at rank r begins with the list of length r Egyptian fractions representing the possible

$$\sum_V \sum_{i=1}^{n_V^2} \frac{1}{n_V f_V} = 1,$$

where the (f_V) are the formal codegrees of the Drinfeld ring; see §2.2. Under the Drinfeld assumption, each formal codegree f_V divides f_1 . Moreover, the Drinfeld ring is commutative if and only if $n_V = 1$ for all V . In this case, we can restrict to Egyptian fractions of length r satisfying the divisibility condition $f_V \mid f_1$ for all V .

In the noncommutative setting, it may happen that $n_V f_V$ does not divide f_1 for some V . However, as verified in Appendix C, no such exceptions occur up to rank 8 (but they do appear at rank 9; see §2.2). Therefore, for ranks $r \leq 8$, it is safe to restrict to Egyptian fractions of length r that satisfy the divisibility condition.

At rank 9, Lemma 9.3 shows that the complexified noncommutative Drinfeld ring must be isomorphic to either $\mathbb{C} \oplus M_2(\mathbb{C})^2$ or $\mathbb{C}^5 \oplus M_2(\mathbb{C})$. The exception corresponds to Egyptian fractions of length 5 or 7 with one or two terms having $n_V = 2$, and at least one violating the divisibility condition $n_V f_V \mid f_1$.

We verified that for $\text{FPdim} \leq 32000$, the values arising from these exceptional cases are already covered by those for the Egyptian fractions of length 9 satisfying the divisibility condition. Details of this computation can be found in the file `InvestNCRank9Except.txt` of the `Data/EgyptianFractionsDiv/Except` folder of [7].

Appendix B. MNSD DRINFELD RINGS

As established in §8, the Grothendieck ring of any odd-dimensional integral fusion category over $\mathbb{C}$ is an MNSD integral Drinfeld ring (Definition 8.6). This section provides a complete classification of such rings up to rank 9.

B.1. Up to rank 5. There are four MNSD integral Drinfeld rings up to rank 5, contained in §A.1, namely the Grothendieck rings of $\text{Rep}(G)$, with $G = C_1, C_3, C_5, C_7 \rtimes C_3$.

B.2. Rank 7. Here is the complete list of 4 MNSD integral Drinfeld ring of rank 7. Copy-pastable data can be found in the file `MNSDRank7DataOnly.txt.txt`, located in the `Data/Odd` directory of [7].

- (1) FPdim 7, type $[1, 1, 1, 1, 1, 1, 1]$, duality $[0, 6, 5, 4, 3, 2, 1]$, fusion data: see §A.3.1.
 (2) FPdim 39, type $[1, 1, 1, 3, 3, 3, 3]$, duality $[0, 2, 1, 6, 5, 4, 3]$, fusion data:

[illegible]

- Formal codegrees: $[3, 3, 13, 13, 13, 13, 39]$,
- Categorification: $\text{Rep}(C_{13} \rtimes C_3)$.

- (3) FPdim 55, type $[1, 1, 1, 1, 1, 5, 5]$, duality $[0, 4, 3, 2, 1, 6, 5]$, fusion data:

[illegible]

- Formal codegrees: $[5, 5, 5, 5, 11, 11, 55]$,
- Categorification: $\text{Rep}(C_{11} \rtimes C_5)$.

- (4) FPdim 903, type $[1, 1, 1, 3, 3, 21, 21]$, duality $[0, 2, 1, 4, 3, 6, 5]$, fusion data:

1 0 0 0 0 0 0 0	0 1 0 0 0 0 0 0	0 0 1 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	1
0 1 0 0 0 0 0 0	0 0 1 0 0 0 0 0	1 0 0 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	0
0 0 1 0 0 0 0 0	1 0 0 0 0 0 0 0	0 1 0 0 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 0 1	1
0 0 0 1 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 1 0 0 0 0	0 0 0 1 2 0 0 0	1 1 1 1 1 0 0 0	1 1 1 1 1 0 0 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	3
0 0 0 0 1 0 0 0	0 0 0 0 1 0 0 0	0 0 0 0 1 0 0 0	1 1 1 1 1 0 0 0	0 0 0 2 1 0 0 0	0 0 0 2 1 0 0 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	3
0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 1 0 0	0 0 0 0 0 3 0 0	0 0 0 0 0 3 0 0	0 0 0 0 0 3 0 0	0 0 0 0 0 3 0 0	0 0 0 0 0 3 0 0	0 0 0 0 0 3 0 0	10
0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 1 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	0 0 0 0 0 0 3 0	1 1 1 3 3 0 10	1 1 1 3 3 0 10	0 0 0 0 0 11 10	10
0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 1	0 0 0 0 0 0 0 3	0 0 0 0 0 0 0 3	1 1 1 3 3 0 10	1 1 1 3 3 0 10	0 0 0 0 0 11 10	10

- Formal codegrees: $[3, 3, 7, 7, 43, 43, 903]$,
- Property: extension of $\text{ch}(C_7 \rtimes C_3)$,
- Categorification: excluded by §6.4.

B.3. Rank 9. Among the MNSD Drinfeld rings of rank 9, there are 10 that are 1-Frobenius, and 2 that are neither perfect nor 1-Frobenius. Copy-pastable data can be found in the files `MNSD1FrobRank9DataOnly.txt` and `N1FrobMNSDRank9.txt`, located in the `Data/odd` directory of [7].

B.3.1. *1-Frobenius case.* Here is the complete list of 10 MNSD integral 1-Frobenius Drinfeld ring of rank 9:

- (1) FPdim 9, type $[1, 1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 8, 7, 6, 5, 4, 3, 2, 1]$, fusion data:

001000000000	010000000000	001000000000	000100000000	000010000000	000000100000	000000010000	000000001000	000000000100	000000000010	000000000001
001000000000	000000000001	000000000100	000000000100	000000000100	000010000000	000100000000	000000100000	000000100000	000010000000	100000000000
001000000000	000000000100	000000000100	000000000100	000000000100	000000000000	000010000000	000001000000	000001000000	100000000000	000000100000
000100000000	000000000100	000000000100	000000000100	000000000100	000100000000	000000000000	000000000000	100000000000	000000100000	000000100000
000001000000	000100000000	010000000000	001000000000	000000100000	000000100000	100000000000	000000000000	000000000001	000000001000	000000000100
000000000100	000001000000	000001000000	100000000000	000000000000	000000000001	000000000000	000100000000	000100000000	001000000000	001000000000
000000000100	000001000000	100000000000	000000000000	000000000100	000000000100	000000000000	000000000000	001000000000	001000000000	001000000000
000000000001	100000000000	000000000000	000001000000	000000000100	000000000100	000000000100	000000000100	001000000000	000100000000	010000000000

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed
- Categorification: $\text{Vec}(C_3^2)$.

- (2) FPdim 9, type $[1, 1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 8, 7, 6, 5, 4, 3, 2, 1]$, fusion data:

[illegible]

- Formal codegrees: $[9, 9, 9, 9, 9, 9, 9, 9, 9]$,
- Property: pointed,
- Categorification: $\text{Vec}(C_9)$.

- (3) FPdim 57, type $[1, 1, 1, 3, 3, 3, 3, 3, 3]$, duality $[0, 2, 1, 8, 7, 6, 5, 4, 3]$, fusion data:

001000000000	001000000000	001000000000	000010000000	000001000000	000000010000	000000001000	000000000100	000000000010	000000000001
001000000000	001000000000	100000000000	000010000000	000001000000	000000010000	000000001000	000000000100	000000000010	000000000001
001000000000	100000000000	010000000000	000010000000	000001000000	000000010000	000000001000	000000000100	000000000010	000000000001
000010000000	000010000000	000010000000	000000000012	00000100101	000001100101	000001100100	000011100100	00000001110	11100011100
000001000000	000001000000	000001000000	000000101011	000000010101	000001010101	000001010101	000001001110	111100000001	00000111100
000000100000	000000100000	000000100000	000001100100	000001010101	000010100200	11101000100	000001100010	000001100010	00000001111
000000001000	000000001000	000000001000	000000010010	000000010010	000000010010	000000010010	000000010010	000000010010	000000010010
000000000100	000000000100	000000000100	000000001110	111100000110	00000110011	00000110011	000001000101	00000201000	00001010010
000000000001	000000000001	000000000001	11100011100	0000011100	00000000111	00000000111	000001000101	000001000100	00021000000

- Formal codegrees: $[3, 3, 19, 19, 19, 19, 19, 19, 57]$,
- Categorification: $\text{Rep}(C_{19} \rtimes C_3)$.

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix}, \quad \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 & 1 \end{bmatrix}$$

- Formal codegrees: $[3_2, 4, 15, 60]$,
- Property: noncommutative, single possible induction matrix
- Categorification: $\mathcal{C}(A_5, 1, A_4, 1)$, group-theoretical, Morita equivalent to $\text{Vec}(A_5)$, see §6.2.

C.3.1. *Non-1-Frobenius.* Here are the 5 noncommutative non-1-Frobenius integral Drinfeld rings of rank 8:

[illegible]

- [illegible]

- [illegible]

- [illegible]

- [illegible]

- Formal codegrees: $[3_2, 5, 8, 123, 4920]$,
- Property: noncommutative, non-1-Frobenius,
- Categorification: excluded by the fusion subring of type $[1, 1, 1, 6, 9]$, see §A.1.5 (15).

C.3.2. *1-Frobenius*. Here are the 20 noncommutative 1-Frobenius integral Drinfeld rings of rank 8:

(1) FPdim 8, type $[1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 1, 2, 3, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4_2, 8, 8, 8, 8]$,
- Property: noncommutative,
- Categorification: $\text{Vec}(D_4)$.

(2) FPdim 8, type $[1, 1, 1, 1, 1, 1, 1, 1]$, duality $[0, 1, 7, 6, 5, 4, 3, 2]$, fusion data:

[illegible]

- Formal codegrees: $[4_2, 8, 8, 8, 8]$,
- Property: noncommutative,
- Categorification: $\text{Vec}(Q_8)$.

(3) FPdim 20, type $[1, 1, 1, 1, 2, 2, 2, 2]$, duality $[0, 1, 2, 3, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4, 4, 5_2, 20, 20]$,
- Property: noncommutative,
- Categorification: $\mathcal{C}(F_5, 1, D_5, 1)$, $\mathcal{C}(F_5, 1, C_2, 1)$.

(4) FPdim 20, type $[1, 1, 1, 1, 2, 2, 2, 2]$, duality $[0, 1, 3, 2, 4, 5, 7, 6]$, fusion data:

[illegible]

- Formal codegrees: $[4, 4, 5_2, 20, 20]$,
- Property: noncommutative,
- Categorification:

(5) FPdim 24, type $[1, 1, 1, 1, 1, 1, 3, 3]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 8, 12, 12, 24]$,
- Property: noncommutative,
- Categorification: excluded by Lemma 5.3.

(6) FPdim 24, type $[1, 1, 1, 1, 1, 1, 3, 3]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 8, 8, 24, 24]$,
- Property: noncommutative,
- Categorification: $\mathcal{C}(S_4, 1, A_4, 1), \mathcal{C}(S_4, 1, S_3, 1)$.

- Formal codegrees: $[3_2, 4, 16, 60, 240]$,
- Property: noncommutative,
- Categorification:

(14) FPdim 600, type $[1, 1, 1, 2, 2, 2, 3, 24]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 24, 25, 600]$,
- Property: noncommutative,
- Categorification:

(15) FPdim 816, type $[1, 1, 1, 3, 6, 16, 16, 16]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 16, 51, 816]$,
- Property: noncommutative,
- Categorification:

(16) FPdim 960, type $[1, 1, 1, 3, 4, 4, 4, 30]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 15, 64, 960]$,
- Property: noncommutative,
- Categorification:

(17) FPdim 1806, type $[1, 1, 1, 1, 1, 1, 6, 42]$, duality $[0, 1, 2, 3, 5, 4, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 6, 7, 43, 1806]$,
- Property: noncommutative,
- Categorification:

(18) FPdim 2184, type $[1, 1, 1, 3, 12, 26, 26, 26]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 13, 168, 2184]$,
- Property: noncommutative,
- Categorification:

(19) FPdim 3660, type $[1, 1, 1, 3, 4, 4, 4, 60]$, duality $[0, 2, 1, 3, 4, 5, 6, 7]$, fusion data:

[illegible]

- Formal codegrees: $[3_2, 4, 15, 61, 3660]$,
- Property: noncommutative,
- Categorification:

